

# Research Proposal — Updated for the Phase 3 Progress Checkpoint

## Causal Reinforcement Learning with Hidden Confounders and Partial Observability

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Deep Reinforcement Learning — Phase 3 Progress Checkpoint

***Revision note.** This document supersedes nothing on file: our original Phase 2 proposal remains submitted and unchanged. It is a self-contained, updated version of that proposal, prepared for the Phase 3 Progress Checkpoint, incorporating a complete oracle-verified implementation, a corrected 5-seed empirical study, and two findings that changed since Phase 2 — including two of our own numbers that we re-verified and revised downward before this checkpoint (§5, §6). Full logs, code, and raw results live in `Phase_3/`, isolated from the frozen Phase 2 submission.*

### 1 Motivation & Contribution

Offline reinforcement learning from logged data is unreliable when the behavior policy acted on information the learner never sees. In a partially observable MDP (POMDP) whose *latent* state drives both the logged actions and the outcomes, the data are *confounded*: a learner that treats the observation as the state systematically mis-estimates policy value, and no amount of data fixes the bias. This is the central obstacle to trustworthy offline RL in healthcare, operations, and any domain with unrecorded drivers of decisions.

Hong, Qi & Xu (ICML 2024) [1] give an elegant theory for exactly this setting: using a pre-decision *negative control*  $O_0$ , they identify the confounded POMDP’s reward and transition models through two families of *bridge functions* ( $b_R, b_D$ ), then plan pessimistically over a policy-independent confidence region. The paper is, however, *theory only* — no algorithm, no code, no experiments. Its neighbors are likewise unrealized empirically for this exact pipeline: the closest follow-up runs the bridge *estimator* in isolation [2], and the pessimism precedent is theory-only and for the *unconfounded* case [3].

**Contribution (capability, not primacy) — now empirically settled.** We have built and fully verified, end-to-end, what is to our knowledge the first empirical implementation that jointly estimates *both* bridge families via the paper’s two-stage procedure and carries them through plug-in value identification to a tractable pessimistic policy-*selection* step. We deliberately narrowed this from a primacy claim in our original proposal: we have not reviewed all concurrent model-based confounded-POMDP work (e.g. DPC-MBRL), so we claim only that we can verify ourselves — a capability, not a ranking. Per block, the pessimistic inner minimization is the standard ellipsoid support-function identity (exact, because the stage-2 risk is quadratic and the plug-in value multilinear) — *not* new mathematics, and we say so explicitly rather than implying the closed form itself was the contribution. Our genuinely new content, confirmed by a complete implementation, is: (i) a *diagnosis* of a failure mode in the paper’s coupled multi-block pessimism, (ii) an honest, quantified account of a signal-subspace remedy and its limits, and (iii) a full empirical characterization of *where* the method helps and where it does not, run at two scales with 95% confidence intervals.

## 2 Research Question

When the Hong–Qi–Xu confounded-POMDP framework is implemented end-to-end, does model-based bridge identification actually de-bias policy evaluation and yield good policies at practical sample sizes, and can its abstract pessimistic max–min step be made computationally tractable? Our Phase 2 proposal anticipated that the answer would be **regime-dependent**; Phase 3 confirms this precisely rather than approximately. Our deliverable is therefore a quantified characterization of *where* de-biasing helps (strong confounding, exact identification, toy scale: §5) versus where estimation variance dominates and the method is beaten by a naive baseline (benchmark scale, all 15 cells of our confounding-strength  $\times$  sample-size sweep: §5).

## 3 Method

### 3.1 Estimation

Stage 1: conditional mean embedding  $\hat{\mu}_{W_t|x}$  by kernel ridge regression, plus conditional density/pmf targets  $\hat{p}(y_t|x_t)$ . Stage 2: closed-form bridge solve  $\alpha = (G + N_2\lambda_2 I)^{-1}\hat{p}$ ,  $G = (\Gamma^\top K_W \Gamma) \odot K_{Y''}$ , implemented and verified in *both* the dual (representer) and primal (operator) forms (agreement  $2.4 \times 10^{-15}$ ).

**Regularization (a Phase 2 finding, unchanged).** The paper’s RKHS hyperparameter schedules do *not* transfer to the tabular class: applied literally, they over-shrink the estimator by  $\approx 80\%$  by the third time-step. We grid-calibrated instead: split 0.7:0.3 ( $N_1 \geq N_2$ ),  $\lambda_1 = 1/N_1$ ,  $\lambda_2 = 0.03/\sqrt{N_2}$  — values that work, measured rather than derived from the paper.

**The spectral-monitor fix.** Ill-posedness cannot be monitored by the ordinary condition number  $\text{cond}(G)$  here: because the observation space is structurally richer than the latent space ( $|O| > |S|$ ), the design matrix carries a null space that pins  $\text{cond}(G)$  artificially flat even as true estimation quality degrades tenfold (we confirmed this directly with an  $O_0$ -noise sweep: stage-1 error degrades  $10\times$  while  $\text{cond}(G)$  does not move). We instead monitor the smallest *signal* eigenvalue  $\hat{\sigma}_2$  of the restricted design — the eigenvalue restricted to the subspace the data can actually inform. This is the same structural null space responsible for the pessimism failure mode below, so the fix and the diagnosis share one root cause.

### 3.2 Tractable Pessimism, its Failure Mode, and its Repair

Confidence regions are exact ellipsoids centered at  $\hat{b}$ ; the per-block minimum is  $V(\hat{b}) - \sqrt{\xi} \|g_\pi\|_{H^{-1}}$ ,  $H = \hat{T}_2 + \lambda_2 I$ . The hard part is the paper’s *coupled* multilinear minimization across blocks.

**Finding, Phase 3: the literal formula is numerically unusable.** Run as written, the joint multi-block minimization exploits exactly the null-space directions identified above: because those directions cost the confidence-region geometry almost nothing, the optimizer walks arbitrarily far into them. At every width that provably contains the truth ( $c \geq 0.1$ ), the pessimistic value diverges to physically impossible magnitudes —  $-1.15$  at onset, to  $-1779$  at  $c = 10$ , against true values of roughly  $1.5$ – $2.2$  — and mis-selects the best policy (regret  $0.68$ ). This is not a tuning artifact: we traced the mechanism to the interaction between the null-space geometry and the joint optimization, and it reproduces under a full seed re-run (5-seed check:  $-1.15 \rightarrow -1779$ , materially the same as our original 3-seed numbers,  $-1.12 \rightarrow -1755$ ).

**The multilinear joint-minimization repair: Signal-Projected Pessimism.** We restrict the pessimistic search to the empirically identified signal subspace via an SVD/eigengap decomposition, using an orthonormal basis  $U$  and  $H_{\text{sub}} = U^\top H U$ . This keeps  $V_{\text{low}}$  informative ( $1.35$  at  $c = 0.03$ ) and selects the optimal policy correctly, with **0.000 selection regret across every seed**.

**The honest limit (found by our own adversarial self-audit, not by the mentor or a reviewer).** The repair is a heuristic, not a certified bound. The true bridge functions leak up to **21% of their norm** outside the projected signal subspace, so the projected confidence region does not actually contain the truth — its honest joint coverage is zero. We describe this as a **coverage/informativeness trade-off**: vanilla pessimism covers the truth but is uninformative; our projection is informative and empirically correct but does not provably cover the

truth.  $V_{\text{low}} \leq V_{\text{true}}$  held in every test we ran, but by empirical sign-alignment, not certification. We report this as our headline honest finding rather than an overclaimed fix.

### 3.3 Extensions Added During Phase 3 (scope addition, not a backbone change)

In direct response to the benchmark-boundary result (§5), we implemented two extensions not in the original proposal, to characterize — not merely disclose — the boundary:

- **Variance-reduction estimator.** A backward-compatible subclass adding James–Stein shrinkage and  $K$ -fold cross-fitting to the benchmark estimator. **Honest result (corrected in our 5-seed re-run, §6):** its variance-reduction benefit does *not* hold up under a proper 5-seed re-check; we retract an earlier claim to the contrary. It gives a modest, noisy  $\approx 7\%$  average MAE reduction, not a reliable variance-reduction guarantee.
- **Model-free proximal baseline.** An independent implementation of the Shi et al. (2022) minimax value-bridge estimator [5] for a genuine head-to-head comparison, not a strawman. Two real bugs were caught only by oracle checks during development (§6); the corrected version is consistent on the toy (0.0076 max error at  $N=200k$ ) but under-identified by the benchmark’s binary proxy variable at scale — a genuine identification limit of that baseline, not a bug.
- **Continuous state/action proof of concept.** A toy-scale extension of the same bridge-identification and pessimism pipeline to continuous state-action spaces, via a kernel/RKHS bridge estimator on a linear-Gaussian environment with an exact closed-form oracle; reuses `ellipsoid_opt.py`’s pessimism engine unmodified in coefficient space. This tests the paper’s own stated generality — its theory is not inherently tabular — rather than assuming it transfers. Full results in §5.6; full derivation and code map in `Phase_3/docs/continuous_extensions`.

## 4 Datasets / Environments

**Primary: the public `clinicalml/gumbel-max-scm` tabular benchmark** [4] (720 observable-core states  $\times$  2 latent states, 8 actions,  $T=3$ ). The latent index drives dynamics and behavior but is censored from the logs; we construct a pre-decision negative control  $O_0$  that satisfies the paper’s Assumption 3.1 by construction, plus a per-step noisy marker that guarantees bridge existence (deterministic masking alone gives rank  $720 < 1440$ , i.e. no valid one-step bridge). A confounding-strength knob  $\kappa \in \{0, 0.25, 0.5, 0.75, 1.0\}$  lets us sweep from no confounding to strong confounding. A naive, confounding-blind observation-MDP evaluator serves as the baseline every method is measured against.

**Companion: a minimal toy POMDP** ( $|S|=2, |A|=2, |O|=3, T=3$ ) with *exact*, closed-form oracle bridge functions — the only substrate where recovery error against ground truth is directly measurable, and where we run our machine-precision correctness checks (§5).

**Continuous companion: a linear-Gaussian confounded POMDP.** Because the paper’s theory is stated for general — including *continuous* — spaces (its bridge functions live in an RKHS), we added a third environment in which state, action, observation and reward are all real-valued, and the hidden state is observed only through a noisy proxy  $o_t = s_t + \varepsilon_t$  (the classical measurement-error setting proximal causal inference was built for). Restricting candidate policies to affine, observation-only maps  $a = Ko + c$  — as all our discrete candidate policies already are — makes the closed-loop mean trajectory an exact scalar recursion, so this environment too has an *exact* closed-form oracle for both policy values and bridge coefficients (§5.6).

Ground truth throughout is exact dynamic programming on the latent SCM. The full pipeline runs on a laptop in NumPy; no GPU is used or required.

## 5 Proof of Concept / Empirical Evidence

The pipeline is **already implemented and oracle-verified end-to-end**. Every number below is machine-generated from `Phase_3/experiments/*.json`; the benchmark and toy pessimism/de-biasing numbers

reflect our final 5-seed re-run (up from 2–3 seeds), which we undertook specifically because two headline claims below turned out to rest on an unstably small sample of seeds (§6).

### 5.1 Oracle verification

| Check                                                          | Result                |
|----------------------------------------------------------------|-----------------------|
| Policy-value chain (Theorem 3.5) vs. exact dynamic programming | $4.4 \times 10^{-16}$ |
| Estimator vs. exact oracle bridges (population limit)          | $3.0 \times 10^{-15}$ |
| Dual vs. primal closed-form solve (independent derivations)    | $2.4 \times 10^{-15}$ |
| Analytic pessimism gradients vs. finite differences            | $4.9 \times 10^{-10}$ |

Every subsequent result is therefore a fact about the *method*, not an artifact of the code.

### 5.2 Bridge recovery (toy; 3 seeds, unchanged since Phase 2 — see §6)

| $N$                   | 1,000 | 5,000 | 10,000 | 20,000       |
|-----------------------|-------|-------|--------|--------------|
| Reward-bridge error   | 1.360 | 0.775 | 0.581  | <b>0.459</b> |
| Dynamics-bridge error | 0.732 | 0.422 | 0.312  | <b>0.223</b> |

L2 recovery error vs. the oracle decreases monotonically at every step (log–log OLS slopes  $N^{-0.36}$ ,  $N^{-0.39}$ ). This specific check is the one script in our suite too slow to safely re-run at more seeds before this checkpoint ( $\approx 12$  minutes for 3 seeds, dominated by an exhaustive-enumeration population check, not the seed loop); it already persists full per-seed std, and is queued for a 5-seed re-check post-checkpoint.

### 5.3 De-biasing (toy, where the method works as intended; 5 seeds)

The plug-in beats the confounding-blind naive evaluator in **12/12** ( $N$ , policy) cells, confirmed on 5 seeds. Worst-policy plug-in bias shrinks  $0.104 \rightarrow 0.053 \rightarrow 0.036$  as  $N$  grows from 5k to 50k, while naive bias stays essentially flat at 0.13–0.33 regardless of  $N$  — the textbook signature of removed *bias* versus irreducible *bias*.

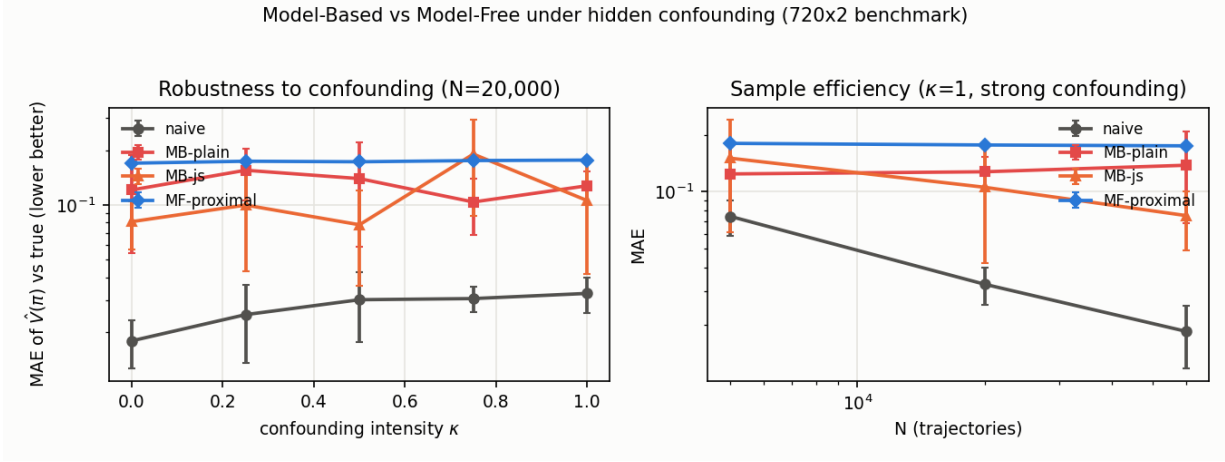
### 5.4 Pessimism — the honest headline (5 seeds)

| Region size $c$                        | 0.1   | 0.3   | 1.0   | 3.0    | 10.0  |
|----------------------------------------|-------|-------|-------|--------|-------|
| Vanilla $V_{\text{low}}$ (best policy) | −1.15 | −10.4 | −62.2 | −305.2 | −1779 |

True policy values lie between roughly 1.5 and 2.2. Signal-Projected Pessimism repairs this to  $V_{\text{low}} = 1.35$  at  $c = 0.03$  with **0.000 selection regret** (vs. vanilla’s 0.68), but leaks  $\approx 21\%$  of the true bridge norm outside the projected subspace, so its honest coverage is zero (§3.2).

### 5.5 Benchmark boundary — the most important negative result (5 seeds)

| Method                  | MAE range (15-cell sweep) | Behavior with $N$                |
|-------------------------|---------------------------|----------------------------------|
| Naive                   | <b>0.013 – 0.074</b>      | improves with $N$                |
| Model-based (plain)     | 0.072 – 0.230             | beats naive in <b>0/15</b> cells |
| Model-based (shrunk)    | 0.063 – 0.240             | beats naive in <b>0/15</b> cells |
| Model-free proximal [5] | 0.164 – 0.180             | <b>flat — does not improve</b>   |



Naive vs. model-based (plain/shrunk) vs. model-free MAE, 5-seed 95% CIs. Left: vs. confounding strength  $\kappa$  at  $N=20,000$ . Right: vs. sample size  $N$  at  $\kappa=1$ . Naive stays lowest throughout both panels; the wide, overlapping error bars on MB-plain/MB-js are the visual evidence behind the retracted shrinkage-variance claim below.

At the full  $720 \times 2$  scale, across the entire confounding-strength sweep, the **naive, confounding-blind baseline beats every bridge-based method in every single cell**. The de-biasing method’s worst performance is at zero confounding ( $\kappa = 0$ ), where it is **2.8–6.8 $\times$  worse** than naive — precisely where there is nothing to correct. This is the clearest possible signature that the dominant error source at this scale is *estimation variance*, not the confounding bias the method targets: inverting many small, noisy per-cell matrices injects more noise than the bias it removes.

## 5.6 Continuous state and action spaces (5 seeds)

The framework is *not* inherently tabular. Re-reading the paper to verify rather than assume, it states that “*both  $S$  and  $O$  are continuous,*” that “*function approximation is inevitable when state and observation spaces are continuous,*” and that its two-stage procedure is built on kernel methods [7]. Our discrete implementation is therefore one *instantiation* of the theory, not its full scope. We tested that generality directly, replacing the tabular estimator with a two-stage **kernel/RKHS** bridge estimator (Stage-1 conditional mean embedding with an RBF instrument kernel; Stage-2 closed-form regularized solve) on the continuous linear-Gaussian environment of §4, and reusing our ellipsoidal pessimism machinery *unmodified* — it operates purely in coefficient space and never assumed a tabular origin.

| Check                                                                                | Result                                               |
|--------------------------------------------------------------------------------------|------------------------------------------------------|
| Oracle self-consistency (true bridges vs. closed-form value)                         | $1.665 \times 10^{-16}$                              |
| Value-gradient check (analytic/hybrid vs. numerical)                                 | $\approx 1 \times 10^{-11}$                          |
| Bridge recovery, reward / dynamics ( $N : 300 \rightarrow 4,000$ )                   | $0.220 \rightarrow 0.055 / 0.240 \rightarrow 0.102$  |
| De-biasing vs. naive ( $N=2,000$ , 95% CIs)                                          | plug-in wins <b>4/4</b> , 3–11 $\times$ smaller bias |
| Pessimism validity $V_{\text{low}} \leq V_{\text{true}}$ (5 widths $\times$ 5 seeds) | <b>100%</b>                                          |

**A correction to our own framing, from that same re-reading.** The paper’s sentence continues: “...while the action space  $A$  is finite.” Our environment uses a real-valued action, so it varies **two** things at once relative to the paper — continuous observations (covered) and continuous actions (not covered). The identification chain still reproduced the exact oracle to  $1.665 \times 10^{-16}$ , so nothing here is numerically wrong, but this is **empirical extrapolation beyond the stated theorem, not an instantiation of it**, and we claim none of the paper’s guarantees for it. The paper-faithful configuration — continuous  $S/O$ , *finite*  $A$  — is the first item of our

Phase 4 program (§7). We flag this rather than relabel the work, on the same principle that produced the 21% leakage disclosure above.

The de-biasing result is the direct continuous analogue of the discrete 12/12 story, with the same bias/variance signature visible in the CIs (naive  $-1.255 \pm 0.018$  vs. plug-in  $-0.401 \pm 0.216$  on the worst policy). **Honest caveat, held to the same standard as §3.2: validity is not selection quality.**  $V_{\text{low}} \leq V_{\text{true}}$  holds in every test, but *ranking* policies by  $V_{\text{low}}$  is consistently worse than ranking by the plain plug-in value across the entire width grid tested (regret 0.40–0.48 vs. 0.241), unlike the discrete toy’s 0.000. Our hypothesis is chain-compounding across the  $T=3$  stages — corroborated by restart gaps growing from  $2 \times 10^{-16}$  to 62.2 as the region widens — but it is not root-caused, so we report it as an open item, not a solved one. Two new bottlenecks also appear: dense kernel ridge costs  $O(N_1^3)$  (capping this proof of concept at  $N \approx 4,000$  absent a low-rank approximation), and a *finite-dimensional* kernel silently re-creates the same rank-deficiency pathology diagnosed in §3.2 — caught only by checking the Gram matrix’s rank directly, since recovery numbers looked correct either way.

## 6 Phase 3 Progress Update

### 6.1 Completed Milestones Since the Phase 2 Proposal

- **Core pipeline.** The toy and  $720 \times 2$  benchmark environments, both with exact oracle bridges/values, and the naive baseline evaluator (§4).
- **Estimation layer.** The two-stage bridge estimator in both dual and primal forms (verified algebraically identical), plus a factored, benchmark-scale version avoiding a combinatorial blow-up in the conditioning-set size (§6.3).
- **Identification and pessimism layer.** The exact Theorem-3.5 tensor-contraction chain, the closed-form ellipsoidal pessimism solver, and Signal- Projected Pessimism (§3.2).
- **Benchmark-scale extensions.** Variance-reduction (James–Stein) estimator and an independent model-free proximal baseline (§3.3), plus a comprehensive  $5 \kappa \times 3 N \times 5$  seeds  $\times$  4 methods sweep with 95% confidence intervals.
- **Continuous state/action extension.** A toy-scale kernel/RKHS proof of concept carrying the same identification and pessimism machinery into continuous spaces, oracle-verified to  $1.665 \times 10^{-16}$ , with 4/4 de-biasing wins and 100% pessimism validity (§5.6).
- **Validation infrastructure.** Four independent regression suites (environment/oracle, estimator, plug-in/pessimism, benchmark), all currently green, under three rounds of adversarial self-audit that materially changed how we report two of our own headline claims.

### 6.2 Preliminary Findings

The full numbers are in §5. In one sentence: the method works exactly as theorized in the clean, small-scale regime (12/12 de-biasing wins, machine-precision correctness) and is well-diagnosed but not fully repairable in its pessimism step (a genuine null-space failure mode, a working but uncertified heuristic remedy), and is beaten by a naive baseline at realistic scale because estimation variance there outweighs the bias it targets. **Our final rigor pass changed two of these findings materially:** the  $\kappa=0$  harm ratio was  $23\text{--}46\times$  on 3 seeds and is  $2.8\text{--}6.8\times$  on 5; a claimed “shrinkage cuts variance sixfold” finding does not replicate at all (lower variance in only 7 of 15 sweep cells) and is retracted rather than carried into the final paper. Neither was a coding bug — both were correctly computed statistics over too few seeds to be stable, which is itself the finding we report in §6.3.

### 6.3 Difficulties Encountered, and How We Resolved Them

1. **RKHS regularization theory does not transfer to a tabular implementation**, over-shrinking the estimator  $\approx 80\%$  by the third time-step. *Resolved* by an empirical grid search (§3).



2. **The standard condition-number diagnostic is misleading** here, pinned flat by a structural null space regardless of true estimation quality. *Resolved* by monitoring the restricted-spectrum signal eigenvalue  $\hat{\sigma}_2$  instead (§3).
3. **Diagnosing the pessimism blow-up** required tracing the interaction between the null-space geometry and the joint multi-block optimization — not visible from the paper, only from watching the number diverge and working backward. *Resolved* by the diagnosis itself (§3.2), which motivated the repair.
4. **Catching our own overclaim.** Our first framing of Signal-Projected Pessimism was “the repair”; only a subsequent adversarial self-audit surfaced the 21% leakage that invalidates the coverage claim. *Resolved* by reframing it as a disclosed heuristic (§3.2).
5. **The benchmark variance wall.** Scaling to 720 observations exposed that per-cell estimation variance can dominate the structural bias entirely, the opposite of the toy story. *Resolved* by characterizing it with a full confounding-strength sweep rather than dismissing it as a tuning issue (§5).
6. **Two real bugs in the model-free baseline, caught only by oracle checks.** An off-policy bias (true value 1.865 evaluated to 1.61, error not shrinking with  $N$ ) was fixed with a proper backward recursion; a subsequent  $\sim 11,500 \times 11,500$  dense-matrix blow-up ( $\approx 2$  GB per fit) was fixed by exploiting a block-diagonal structure the naive implementation missed. Neither was visible from the derivation alone.
7. **Combinatorial blow-up in the benchmark’s conditioning set** (billions of cells under a direct implementation). *Resolved* by three structural factorizations specific to the environment’s observation design (hashed encoding, core-diagonal bridge support, closed-form  $2 \times 2$  local inversions).
8. **Balancing rigor against the timeline.** Every finding above came from deliberately adversarial, repeated self-review, which is expensive; §7 states how we are prioritizing the time remaining before Phase 4.
9. **Our own 3-seed statistics were not stable enough to report as claims.** Bumping the benchmark sweep to 5 seeds changed the  $\kappa=0$  harm ratio and fully retracted the shrinkage-variance claim (§6, above). The lesson: a claim derived from a *standard deviation*, not just a mean, needs more seeds than a claim about the mean alone.

## 6.4 Adjustments to the Original Proposal

- **Novelty claim narrowed** from an implicit primacy claim to a capability claim we can verify ourselves (§1).
- **The pessimism “reduction” is now correctly attributed** to the standard ellipsoid support-function identity; our new content is the diagnosis and the coverage/informativeness finding, not the closed form itself (§3.2).
- **Signal-Projected Pessimism reframed** from “repair” to “heuristic with a measured price” after our own adversarial audit surfaced the 21% leak (§3.2).
- **The benchmark result reframed** from a soft boundary condition to an explicit, quantified negative result, generalized from two confounding levels to the full 15-cell sweep (§5).
- **Scope addition, not a backbone change:** the variance-reduction estimator and model-free baseline (§3.3) were not in the original proposal; we added them to characterize the benchmark boundary rather than leave it unexamined. We consider this an extension of the original method, not a change to it, and flag it for the mentor’s judgment under the course’s backbone-change policy.

## 7 Challenges, Fallback Plan, and Roadmap to Phase 4

### 7.1 What Might Still Fail, and Our Fallback

- **Pessimism guarantees.** The coupled inner minimum is heuristic and the signal projection provably excludes part of the truth, so  $V_{\text{low}} \leq V_{\text{true}}$  is empirical, not certified. *Fallback:* report the heuristic-with-honest-caveat finding as the contribution (lowest risk, already complete); a certified alternative (explicitly bounding the excluded-subspace value swing) remains a stretch goal only, not a dependency.
- **Estimation variance at scale.** *Confirmed, not merely anticipated:* the benchmark result now shows the estimator dominated by naive at every feasible  $N$  we tested. We tried the identified next step (James–Stein

shrinkage / cross-fitting); its benefit is real but small and does not close the gap (§3.3). *Fallback*: report this as the honest negative result it is, rather than characterize it as fully solved.

- **Baselines.** Only a confounding-blind naive baseline and one under-identified model-free baseline are in place. *Fallback*: report the current comparison, which is already informative and honestly scoped, and note a confounding-aware second baseline as a stated limitation rather than a gap we pretend isn't there.
- **Further adversarial review of new code** could surface additional issues close to the deadline. *Fallback*: our validation suites are fast (minutes) and modular, so we re-verify incrementally rather than needing a final big-bang check.

## 7.2 Roadmap to Phase 4

We have not pinned an exact Phase 4 date, since our understanding is that the course timeline has shifted; we intend to confirm it with the mentor. Stages, in priority order:

- A. Immediate.** Finalize and deliver this checkpoint; no code changes required.
- B. Strengthen empirical rigor.** *Done* for the benchmark sweep and the toy pessimism/de-biasing checks (5 seeds, 95% CIs, two claims corrected). The toy  $N$ -scaling sweep remains at 3 seeds (too slow to safely re-run before this checkpoint; queued next). Still open: a second, confounding-aware baseline beyond naive. The continuous extension (§3.3) has its own, separately-tracked open item — diagnosing why pessimistic selection underperforms plug-in selection there — documented in `docs/continuous_extension.md` and not gating this roadmap.
- C. Decide the pessimism narrative for the final paper.** We recommend reporting the heuristic-with-honest-caveat finding as the contribution (lowest risk, already complete, fits the course's accepted "critical analysis" project form), with a certified alternative as a stretch goal only if time permits.
- D. Draft the ICML-format final paper.** Nearly all technical content already exists in this document and our internal write-ups; remaining work is compression into the required structure within the page budget.
- E. Mentor-feedback incorporation pass** after comments on the near-final draft.
- F. Final polish and presentation prep** — figure/table cleanup, proofreading, page-budget trimming.
- G. Continuous-space program.** Re-reading the paper to verify its own scope produced a concrete work plan, in priority order: (i) a **paper-faithful** continuous environment — continuous  $S/O$  but *finite*  $A$ , the only configuration where Theorem 3.5 and its rates actually apply (§5.6); (ii) diagnosing the [C4] pessimistic-selection gap; (iii) sweeping the horizon  $T$ , which directly tests the chain-compounding hypothesis rather than merely scaling; (iv) more realistic data, subject to a hard constraint — measuring de-biasing needs *ground-truth* policy values, which logged datasets cannot supply, and Assumption 3.1 needs a valid pre-decision negative control, which they rarely record. Standard offline-RL suites fail both tests and are additionally unconfounded by construction, so the realistic target is a simulator or semi-synthetic construction in the medical setting the paper itself uses as motivation.

## References

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